

1. Fetch all employees with their department names.

```
SELECT e.firstName, e.lastName, d.dept_name  
  
FROM employees e  
  
JOIN departments d ON e.department_id = d.dept_id;
```

2. List all customers and their orders.

```
SELECT c.customerName, o.orderNumber  
  
FROM customers c  
  
JOIN orders o ON c.customerNumber = o.customerNumber;
```

3. List all products with their order quantities.

```
SELECT p.productName, od.quantityOrdered  
  
FROM products p  
  
JOIN orderdetails od ON p.productCode = od.productCode;
```

4. Get all employees and their managers' names.

```
SELECT e.firstName AS Employee, m.firstName AS Manager
FROM employees e
LEFT JOIN employees m ON e.managerID = m.employeeNumber;
```

5. List orders with customer names, including those without orders.

```
SELECT c.customerName, o.orderNumber
FROM customers c
LEFT JOIN orders o ON c.customerNumber = o.customerNumber;
```

6. List all employees even if they don't belong to a department.

```
SELECT e.firstName, d.dept_name
FROM employees e
LEFT JOIN departments d ON e.department_id = d.dept_id;
```

7. List all departments and employees, even if the department has no employees.

```
SELECT d.dept_name, e.firstName
FROM departments d
LEFT JOIN employees e ON e.department_id = d.dept_id;
```

8. Find orders with and without shipped dates.

```
SELECT o.orderNumber, o.orderDate, o.shippedDate  
  
FROM orders o  
  
LEFT JOIN shippings s ON o.orderNumber = s.orderNumber;
```

9. Get products that have never been ordered.

```
SELECT p.productName  
  
FROM products p  
  
LEFT JOIN orderdetails od ON p.productCode = od.productCode  
  
WHERE od.productCode IS NULL;
```

10. Get customers who never placed an order.

```
SELECT c.customerName  
  
FROM customers c  
  
LEFT JOIN orders o ON c.customerNumber = o.customerNumber  
  
WHERE o.orderNumber IS NULL;
```

11. Find all products and their categories.

```
SELECT p.productName, pl.productLine  
FROM products p  
JOIN productlines pl ON p.productLine = pl.productLine;
```

12. Show customers along with their payment details.

```
SELECT c.customerName, p.amount, p.paymentDate  
FROM customers c  
JOIN payments p ON c.customerNumber = p.customerNumber;
```

13. Display employees and the office they belong to.

```
SELECT e.firstName, o.city  
FROM employees e  
JOIN offices o ON e.officeCode = o.officeCode;
```

14. Show all orders with product names and quantity.

```
SELECT o.orderNumber, p.productName, od.quantityOrdered  
FROM orders o  
JOIN orderdetails od ON o.orderNumber = od.orderNumber  
JOIN products p ON od.productCode = p.productCode;
```

15. List customers, their orders, and order status.

```
SELECT c.customerName, o.orderNumber, o.status  
FROM customers c
```

JOIN orders o ON c.customerNumber = o.customerNumber;

16. Get a list of employees who are not managers.

```
SELECT e.firstName
FROM employees e
LEFT JOIN employees m ON e.employeeNumber = m.managerID
WHERE m.managerID IS NULL;
```

17. Show the order number and employee who handled the order.

```
SELECT o.orderNumber, e.firstName
FROM orders o
JOIN customers c ON o.customerNumber = c.customerNumber
JOIN employees e ON c.salesRepEmployeeNumber = e.employeeNumber;
```

18. List employees and their reporting managers (full names).

```
SELECT CONCAT(e.firstName, ' ', e.lastName) AS Employee,
       CONCAT(m.firstName, ' ', m.lastName) AS Manager
FROM employees e
LEFT JOIN employees m ON e.managerID = m.employeeNumber;
```

19. List products and their total sales amount.

```
SELECT p.productName, SUM(od.quantityOrdered * od.priceEach) AS totalSales
FROM products p
```

JOIN orderdetails od ON p.productCode = od.productCode

GROUP BY p.productName;

20. List customers with total number of orders.

SELECT c.customerName, COUNT(o.orderNumber) AS totalOrders

FROM customers c

LEFT JOIN orders o ON c.customerNumber = o.customerNumber

GROUP BY c.customerName;

Last 10 Questions (21–30)

21. Display employees who work in offices located in the USA.

SELECT e.firstName, o.country

FROM employees e

JOIN offices o ON e.officeCode = o.officeCode

WHERE o.country = 'USA';

22. Get departments with no employees.

SELECT d.dept_name

FROM departments d

LEFT JOIN employees e ON d.dept_id = e.department_id

WHERE e.employeeNumber IS NULL;

23. Find customers with more than 2 orders.

```
SELECT c.customerName, COUNT(o.orderNumber) AS orderCount
FROM customers c
JOIN orders o ON c.customerNumber = o.customerNumber
GROUP BY c.customerName
HAVING COUNT(o.orderNumber) > 2;
```

24. Show products that are part of an order with status 'Shipped'.

```
SELECT DISTINCT p.productName
FROM products p
JOIN orderdetails od ON p.productCode = od.productCode
JOIN orders o ON od.orderNumber = o.orderNumber
WHERE o.status = 'Shipped';
```

25. Get customers who made payments over \$5000.

```
SELECT c.customerName, p.amount
FROM customers c
JOIN payments p ON c.customerNumber = p.customerNumber
WHERE p.amount > 5000;
```

26. List employees and the number of customers they manage.

```
SELECT e.firstName, COUNT(c.customerNumber) AS customerCount
FROM employees e
```

```
LEFT JOIN customers c ON e.employeeNumber = c.salesRepEmployeeNumber  
GROUP BY e.firstName;
```

27. List orders and the number of products in each order.

```
SELECT o.orderNumber, COUNT(od.productCode) AS totalProducts  
FROM orders o  
JOIN orderdetails od ON o.orderNumber = od.orderNumber  
GROUP BY o.orderNumber;
```

28. Show employees who have not made any sales.

```
SELECT e.firstName  
FROM employees e  
LEFT JOIN customers c ON e.employeeNumber = c.salesRepEmployeeNumber  
WHERE c.customerNumber IS NULL;
```

29. Get the top 3 customers by total payment amount.

```
SELECT c.customerName, SUM(p.amount) AS totalPaid  
FROM customers c  
JOIN payments p ON c.customerNumber = p.customerNumber  
GROUP BY c.customerName  
ORDER BY totalPaid DESC  
LIMIT 3;
```

30. Show the total revenue generated per product line.

```
SELECT pl.productLine, SUM(od.quantityOrdered * od.priceEach) AS revenue
FROM productlines pl
JOIN products p ON pl.productLine = p.productLine
JOIN orderdetails od ON p.productCode = od.productCode
GROUP BY pl.productLine;
```